Gautam Sharma

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B.Tech. - Computer Science and Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	8.73 (Current)	2021 - Present
B.Tech. Minor	Indian Institute of Technology, Guwahati	7.00 (Current)	2022 - Present
Senior Secondary	CBSE Board	95.20%	2021
Secondary	CBSE Board	95.80%	2019

Projects

- **Quantum Powered Business Rule Engine** *Dec 2023*
  *Group Project, **Gold Medal, 12th Inter IIT Tech Meet**, IIT Madras* bit.ly/3tCRDCf
  - Built a business rule engine using **Quantum Computing**, employing DWave's **dwave-system** which makes the process of re-accommodating passengers affected by flight delays/cancellation **easier and faster over classical methods**.
  - Solved the famous **Passenger Recovery Problem** by formulating a mixed integer programme.
  - For preprocessing, used **Prune and Search**. and **Multithreading**. Used **Dynamic Programming** to speed up the process.
  - Solved the problem classically using **gurobipy**, and as a **CQM**, and **BQM** using **Quantum Annealing**.
  - For post-processing, used **Flow Network Algorithms**, and displayed all the solutions using **Streamlit**.

Achievements

- **ICPC Mathura Prelims 2023**: Achieved **AIR 163** (**6th in IIT Guwahati**, all batches combined) among 1000+ teams.
- **Codeforces**: Achieved maximum rating of **1667 (Expert)** (Handle: *vimruLover*)
  - Codeforces Round 881: Secured **Global Rank 90** among 28,000+ participants. (Handle: *g-s01)*
  - Codeforces Round 874: Secured **Global Rank 236** among 30,600+ participants.
- **AtCoder**: Achieved maximum rating of **1102 (5 Kyu)** (Handle: *vimruLover*)
  - AtCoder Beginner Contest 308: Secured **Global Rank 634** among 10,000+ participants.
- **12th Inter IIT Tech Meet**: Secured **Rank 1** among 100+ students from **20 IITs** in the **Quantum Computing Challenge**.
- **APAC Terminal 2023**:Secured **Rank 4** at **Asia-Pacific Terminal** by **Citadel Securities** and **Correlation One**.
- **Abstractify-2023**: Secured **Rank 3** among participants from **top 8 IITs** in the Quant Winter Trading School, **IIT Delhi**.
- **JEE Advanced 2021**: Secured **AIR 530 among 0.15 million students** across the country.
- **JEE Mains 2021**: Secured **AIR 359 among 0.94 million students** across the country.
- **KVPY SX 2021**: Secured **AIR 397** and the KVPY scholarship given to **top 2000** students across the country.

Technical Skills

- **Programming**: C, C++, Python*, JavaScript
- **Web Technologies**: React.js*, Node.js*, HTML, CSS
- **Optimization frameworks**: Gurobi, OR Tools
- **Database Management**: SQL, MongoDB, Firebase*
- **Quantum Computing**: Qiskit, DWave
- **Miscellaneous**: LaTeX, Figma, Markdown, Bash*

** Elementary proficiency*

Key courses taken

- **Computer Science**: Data Structures & Algorithms, Computer Architecture & Organization, Database Management Systems, Computer Networks, Operating Systems, Formal Methods for System Verification
- **Mathematics**: Linear Algebra, Number Theory, Probability & Random Processes, Discrete Mathematics

Positions of Responsibility

- **Competitive Programming Head**: Coding Club, IIT Guwahati *Apr 2023 - Ongoing*